WHOLESALE TRADING & BUSINESS ACCOUNTS ANALYSIS

Using Microsoft Power BI – Mini Project Report

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Course: B.Com (FinTech & AI), 2nd Year

Tool Used: Microsoft Power BI Desktop

1. Introduction

For this project, I developed a Wholesale Trading & Business Accounts Analysis Dashboard using Microsoft Power BI. I selected this topic because sales analysis, customer collections, receivables management and profitability analysis are important aspects of business accounting and financial management.

Wholesale distribution businesses handle multiple transactions involving product sales, customer invoices, discounts, payments and sales returns. Monitoring these activities is important for understanding business performance and maintaining effective financial management.

The main purpose of this dashboard is to transform business transaction data into an interactive visual report that makes it easier to understand sales performance, product profitability, customer payment behaviour, outstanding receivables and branch performance.

The dashboard includes KPI cards, charts, tables and interactive filters to examine different aspects of business operations. It also provides drill-down, drill-through and tooltip functionalities to explore information at different levels of detail.

This project helped me understand how Microsoft Power BI can be used in accounting and business analysis to convert transaction-level data into meaningful information that supports management decision-making.

2. Objectives

The main objectives of this project were:

1. To analyse overall sales and financial performance using Microsoft Power BI.
2. To identify key performance indicators such as total sales, gross profit, profit margin and total collections.
3. To examine product-wise and category-wise sales performance.
4. To understand the impact of discounts on gross profit.
5. To analyse customer sales contributions and outstanding receivables.
6. To identify overdue invoices through ageing analysis.
7. To examine customer collections and payment methods.
8. To analyse sales returns and identify the major reasons for returns.
9. To compare branch, regional and salesperson performance.
10. To create an interactive dashboard using slicers, filters, drill-down, drill-through and tooltips.
11. To understand how business intelligence tools support financial analysis and business decision-making.

3. Dataset Description

The dataset used in this project relates to wholesale trading and business account transactions. It contains information about invoices, invoice line items, customer payments, sales returns and different business dimensions.

The dataset consists of the following major components:

1. FactInvoices: Contains invoice-related information such as invoice number, invoice date, due date, customer details, salesperson, branch, credit terms, invoice amount and payment status.

2. FactInvoiceLines: Contains product-level transaction details, including product name, quantity, selling price, discounts, net sales, cost of goods sold and gross profit.

3. FactPayments: Contains customer payment transactions, payment dates, payment amounts and payment methods. It supports the analysis of customer collections.

4. FactSalesReturns: Contains information about returned products, return quantities, return dates, return reasons and return amounts.

5. Dimension Tables: Include Date, Customer, Product, Salesperson, Branch, City, State and Region information. These tables help categorise and analyse business transactions from different perspectives.

The dataset contains 1,800 invoices, 5,410 invoice lines, 1,884 payment records, 199 return records, 250 customers and 21 products.

These data components provide a foundation for analysing sales performance, profitability, collections, receivables and operational efficiency.

4. Data Preparation

Before analysing the business data through visualisations, data preparation was carried out to ensure that the dataset was suitable for reporting.

The following preparation activities were considered:

1. **Importing the Dataset:** The CSV files containing invoice, invoice-line, payment, return and dimension information were imported into Microsoft Power BI Desktop.
2. **Checking Column Names:** The available columns were examined to understand the information contained in each table.
3. **Checking Data Types:** Date fields, numerical fields and categorical fields were checked to ensure that they had appropriate data types.
4. **Handling Missing Values:** Important columns were checked for missing or blank values to maintain consistency in the analysis.
5. **Creating Relationships:** Relationships were established between invoice headers, invoice lines, payments, returns and the relevant dimension tables.
6. **Date Table:** A shared date table was used to support time-based analysis of invoice transactions. Separate date relationships were considered for payment and return transactions.

7. **Financial Data Validation:** Sales amounts, discount values, cost of goods sold, gross profit and payment amounts were checked for appropriate numerical representation.
8. **Creating DAX Measures:** Measures were created to calculate important financial indicators used in KPI cards, charts and tables.

These preparation steps help establish a consistent foundation for analysing business transactions and presenting financial information through Power BI.

5. DAX Measures

DAX (Data Analysis Expressions) is the formula language used in Power BI to perform calculations and create measures.

In this project, DAX measures are used to calculate important business and financial indicators.

The major measures used or required for the analysis include:

5.1 Total Sales

Calculates the total net sales generated from product transactions after considering applicable discounts. It helps evaluate the overall revenue generated by the business.

5.2 Gross Sales

Represents the total sales value before deducting discounts. It helps compare the original selling value with the actual net sales.

5.3 Total Discount

Calculates the total discount amount offered on product sales. It helps understand the extent of discounts provided to customers.

5.4 Cost of Goods Sold

Represents the total cost associated with the products sold during the reporting period.

5.5 Gross Profit

Calculates the difference between net sales and the cost of goods sold. It indicates the profit generated from trading activities before other operating expenses.

5.6 Gross Profit Margin

Measures gross profit as a percentage of net sales. It helps evaluate how efficiently the business generates profit from its sales.

5.7 Total Invoices

Displays the total number of unique invoices recorded during the reporting period.

5.8 Total Collections

Calculates the total payments received from customers. It helps monitor the cash collected against customer transactions.

5.9 Outstanding Receivables

Represents the amount yet to be collected from customers after considering payments received and accepted return or credit adjustments, where applicable.

5.10 Sales Returns

Calculates the total value of returned products during the selected period.

5.11 Average Invoice Value

Represents the average sales value generated per invoice. It helps understand the average size of customer transactions.

5.12 Units Sold

Calculates the total quantity of products sold during the reporting period.

These measures support the financial calculations and business comparisons presented throughout the dashboard. The exact measure definitions and calculation logic should be verified against the final PBIX file.

6. Data Components and Visuals Used

After preparing the dataset, different visual components were used to analyse sales, profitability, collections and business performance. Each visual provides a different perspective on the available business information.

6.1 Executive Sales Overview

The Executive Sales Overview provides a summary of the overall financial performance of the wholesale business.

The dashboard includes six KPI cards:

- **Total Sales:** Displays the total net sales generated by the business.
- **Gross Profit:** Displays the profit earned after deducting the cost of goods sold.
- **Profit Margin:** Represents gross profit as a percentage of net sales.
- **Total Invoices:** Displays the number of unique invoices recorded.
- **Total Collections:** Shows the total payments received from customers.
- **Outstanding Receivables:** Displays the amount pending collection from customers.

These KPI cards provide management with a quick understanding of the company's financial position and sales performance without examining individual transactions.

6.2 Product & Discount Analysis

The Product & Discount Analysis page focuses on evaluating product performance, sales contribution and discount patterns.

Clustered Column Chart – Net Sales by Category

This chart compares net sales generated by different product categories. It helps identify the categories contributing the highest and lowest sales.

Treemap – Sales by Subcategory

The treemap represents sales contributions across different product subcategories. The size of each section indicates its relative contribution to total sales.

Bar Chart – Gross Profit by Product

This chart compares the gross profit generated by individual products. It helps identify products contributing more significantly to the company's profitability.

Discount Analysis by Category

This analysis examines discounts offered across product categories and helps understand how discounting affects net sales and gross profit.

Together, these visuals help management evaluate product performance and identify areas where pricing and discount policies may require attention.

6.3 Customer & Receivables

The Customer & Receivables page focuses on customer contribution, payment performance and outstanding balances.

Top Customers by Sales

This visual identifies customers contributing the highest sales to the business. It helps understand customer concentration and major revenue contributors.

Customer Outstanding Balance Table

This table presents customer-wise outstanding receivables. It helps identify customers with pending payments and supports collection monitoring.

Overdue Invoice Ageing

The overdue ageing analysis categorises outstanding invoices according to the number of days they remain unpaid.

The ageing categories include:

- Not Due
- 1–30 Days
- 31–60 Days
- 61–90 Days
- 90+ Days

This analysis helps identify overdue balances and understand the extent of delays in customer payments.

6.4 Collections & Returns

The Collections & Returns page examines customer payment transactions and sales return activities.

Line Chart – Monthly Collections

This chart displays customer collections across different months. It helps identify changes in collection performance over time.

Doughnut Chart – Payment Method

This chart represents the distribution of collections according to payment methods. It helps identify the most frequently used payment method or the method contributing the highest collection amount.

Bar Chart – Return Reasons

This visual compares sales returns according to their recorded reasons. It helps identify the major factors contributing to returned products.

Return Amount by Product Category

This analysis compares return amounts across product categories and helps identify categories contributing more significantly to sales returns.

These visuals provide an understanding of payment patterns and potential issues affecting sales and customer satisfaction.

6.5 Branch & Salesperson Performance

This page focuses on evaluating business performance across different geographical locations and sales personnel.

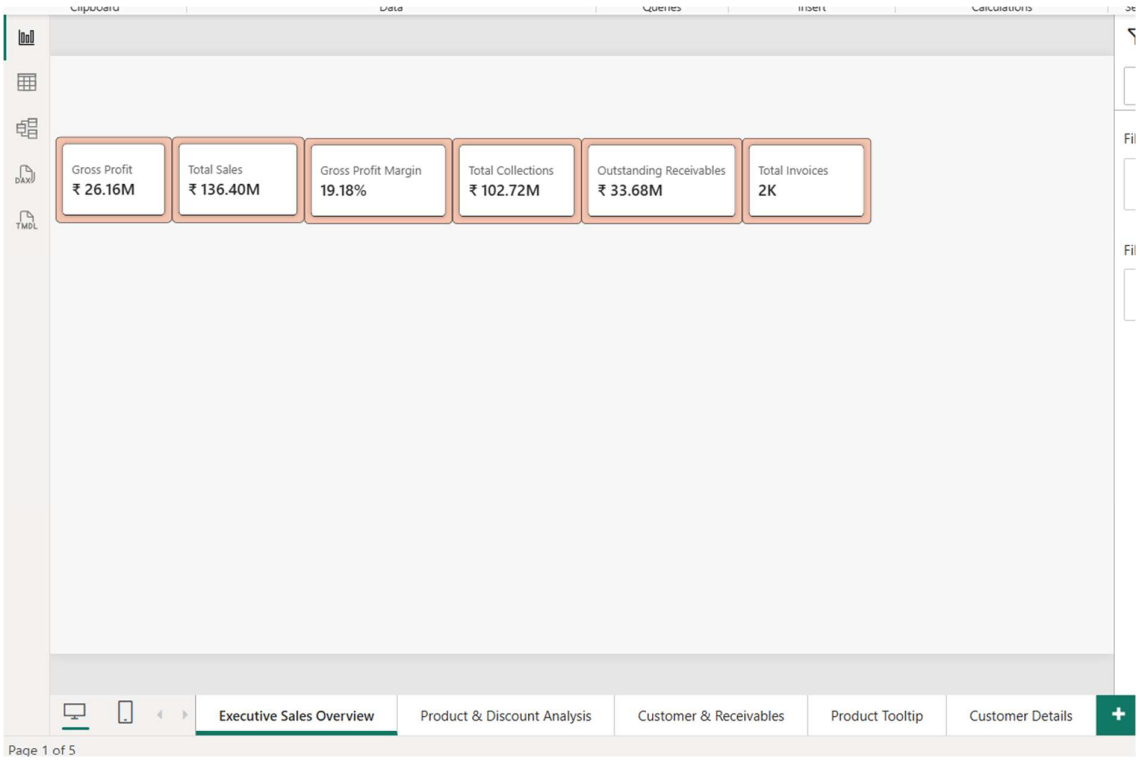
Sales by Region, State and Branch

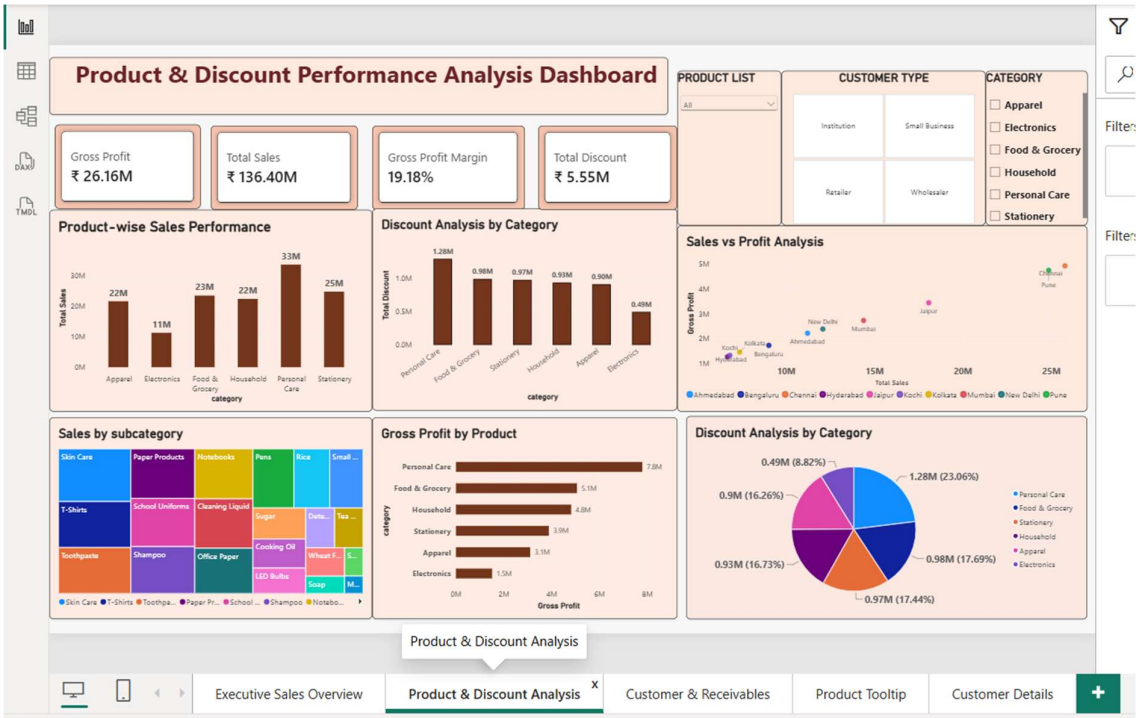
This visual compares sales performance across different geographical locations. It helps identify high-performing and low-performing business areas.

Salesperson Ranking Table

The ranking table compares individual salesperson performance based on total sales, gross profit and invoice count.

This helps management evaluate sales contributions and identify employees with stronger sales performance.




Product & Discount Performance Analysis Dashboard
Gross Profit
₹ 26.16M
Total Sales
₹ 136.40M
Gross Profit Margin
19.18%
Total Discount
₹ 5.55M
PRODUCT LIST
CUSTOMER TYPE
Institution
Small Business
Retailer
Wholesaler
CATEGORY
Apparel
Electronics
Food & Grocery
Household
Personal Care
Stationery
Product-wise Sales Performance
Discount Analysis by Category
Sales vs Profit Analysis
Sales by subcategory
Gross Profit by Product
Discount Analysis by Category
Product & Discount Analysis
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details
Page 2 of 5


Wholesale Trading & Business Accounts Analysis Dashboard
Outstanding Receivables
₹ 33.68M
Total Collections
₹ 102.72M
Total Sales
₹ 136.40M
Total Invoices
2K
CUSTOMER TYPE
Institution
Retailer
Small Business
Wholesaler
MONTH
REGION
PAYMENT METHOD
Bank Transfer
Cheque
Cash
UPI
Filters
Search
Filters on this page
Add data fiel
Filters on all pages
Add data fiel
Customer Outstanding Balance
Monthly Payment Collection Trends
Sales Return Analysis by Reason
Payment Distribution by Method
Top Customers by Sales
Geographical Sales Performance
Customer & Receivables
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details
Page 3 of 5

7. Dashboard Interactivity

Power BI provides interactive functionalities that allow users to explore business information beyond the summary displayed in the charts. In this project, slicers, filters, drill-down, drill-through and report-page tooltips are used to improve the analysis experience.

7.1 Interactive Slicers

Slicers are used to filter the dashboard based on selected business categories.

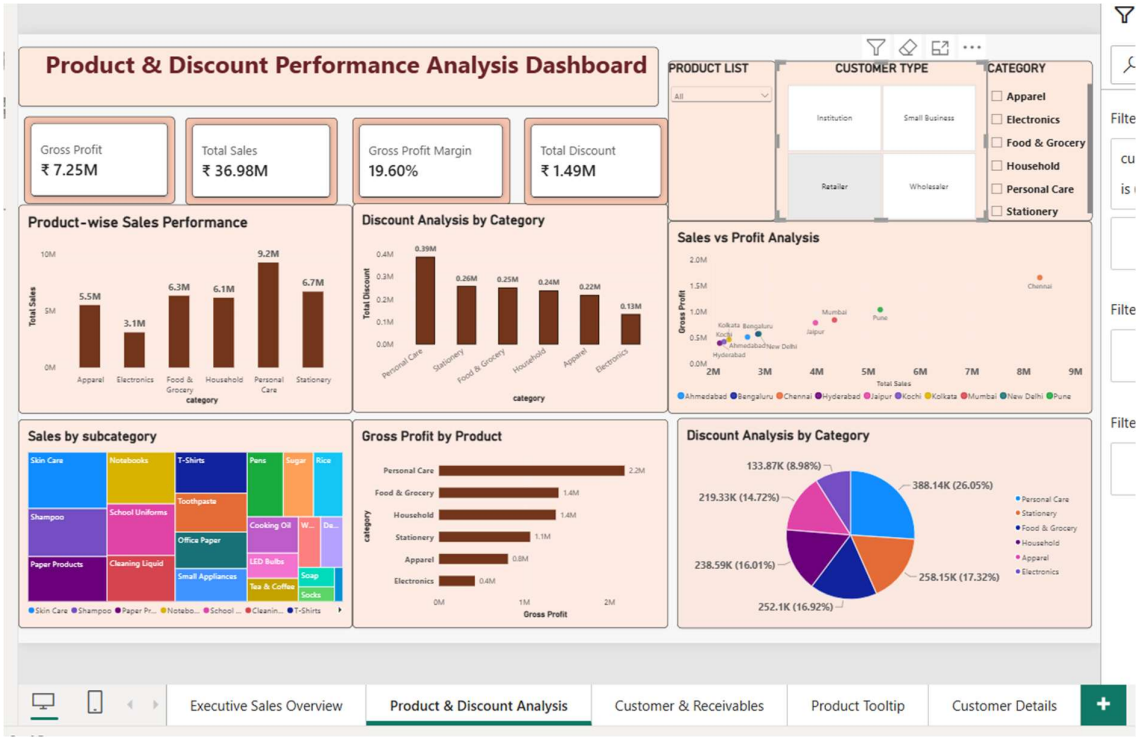
The project includes the following required slicers:

1. Year
2. Region
3. Branch
4. Customer Type
5. Product Category
6. Payment Status

For example, selecting a particular region allows users to examine the corresponding sales and performance information. Similarly, selecting a product category helps analyse the sales and profitability of products within that category.

Slicers make the dashboard flexible and allow users to examine business information according to their requirements.





7.2 Drill-Down Functionality

Drill-down is used to explore information from a higher level of summary to a more detailed level within the same visual.

In this project, two drill-down hierarchies are used.

A. Geographical Hierarchy

Region → State → City → Branch

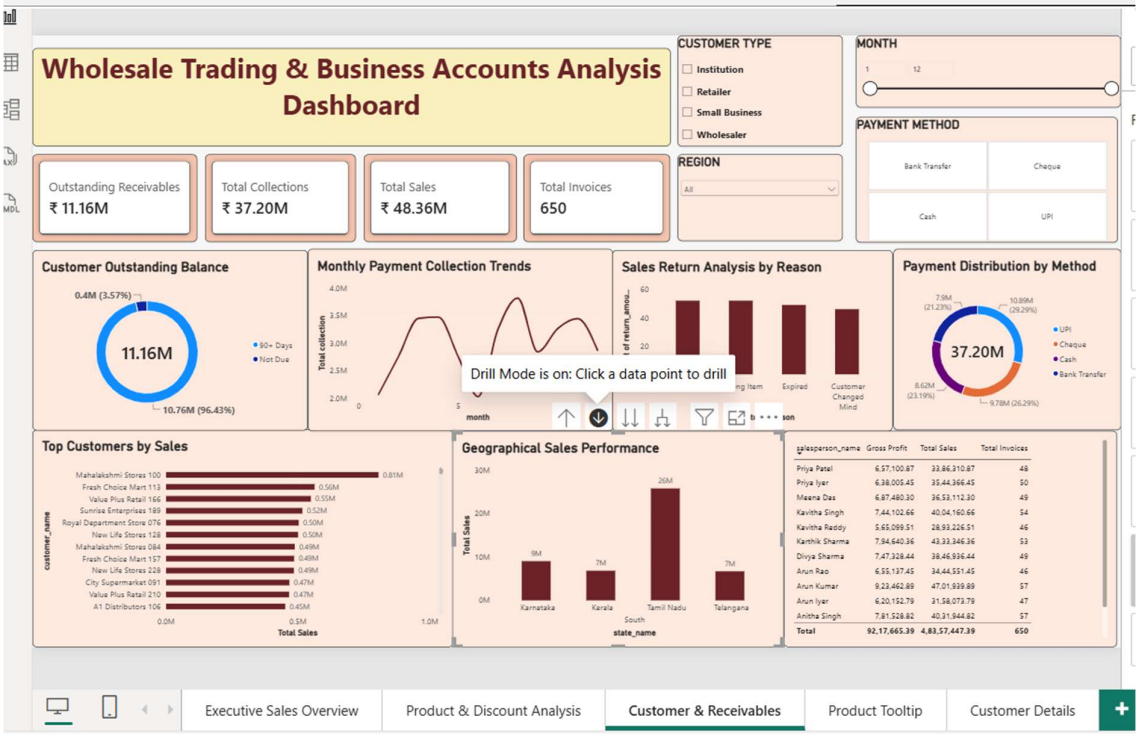
The geographical drill-down allows users to begin with overall regional sales and move to individual states, cities and branches.

For example, when a region is selected, users can explore the sales performance of the states within that region and continue to the city and branch levels.

This helps identify the geographical locations contributing to business sales.


Wholesale Trading & Business Accounts Analysis Dashboard
CUSTOMER TYPE
Institution
Retailer
Small Business
Wholesaler
MONTH
PAYMENT METHOD
Bank Transfer
Cheque
Cash
UPI
REGION
Outstanding Receivables
₹ 33.68M
Total Collections
₹ 102.72M
Total Sales
₹ 136.40M
Total Invoices
2K
Customer Outstanding Balance
33.68M
32.78M (97.32%)
Monthly Payment Collection Trends
Drill Mode is on: Click a data point to drill
Sales Return Analysis by Reason
Payment Distribution by Method
102.72M
Top Customers by Sales
Geographical Sales Performance
East
North
South
West
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details
Filters
Filters on this
branch_name
is (All)
city_name
is (All)
region_name
is (All)
state_name
is (All)
Total Sales
is (All)
Add

3 of 5


Wholesale Trading & Business Accounts Analysis Dashboard
CUSTOMER TYPE
Institution
Retailer
Small Business
Wholesaler
MONTH
PAYMENT METHOD
Bank Transfer
Cheque
Cash
UPI
REGION
Outstanding Receivables
₹ 11.16M
Total Collections
₹ 37.20M
Total Sales
₹ 48.36M
Total Invoices
650
Customer Outstanding Balance
11.16M
10.76M (96.43%)
Monthly Payment Collection Trends
Drill Mode is on: Click a data point to drill
Sales Return Analysis by Reason
Payment Distribution by Method
37.20M
Top Customers by Sales
Geographical Sales Performance
Karnataka
Kerala
Tamil Nadu
Telangana
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details


Wholesale Trading & Business Accounts Analysis Dashboard
CUSTOMER TYPE
Institution
Retailer
Small Business
Wholesaler
MONTH
PAYMENT METHOD
Bank Transfer
Cheque
Cash
UPI
REGION
Outstanding Receivables
₹ 11.16M
Total Collections
₹ 37.20M
Total Sales
₹ 48.36M
Total Invoices
650
Customer Outstanding Balance
11.16M
Monthly Payment Collection Trends
Sales Return Analysis by Reason
Payment Distribution by Method
37.20M
Drill Mode is on: Click a data point to drill
Drill up
Drill down
Total Sales 2,57,88,858.58
Top Customers by Sales
Geographical Sales Performance
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details


Wholesale Trading & Business Accounts Analysis Dashboard
CUSTOMER TYPE
Institution
Retailer
Small Business
Wholesaler
MONTH
PAYMENT METHOD
Bank Transfer
Cheque
Cash
UPI
REGION
Outstanding Receivables
₹ 5.65M
Total Collections
₹ 20.13M
Total Sales
₹ 25.79M
Total Invoices
349
Customer Outstanding Balance
5.65M
Monthly Payment Collection Trends
Sales Return Analysis by Reason
Payment Distribution by Method
20.13M
Drill Mode is on: Click a data point to drill
Top Customers by Sales
Geographical Sales Performance
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details

B. Product Hierarchy

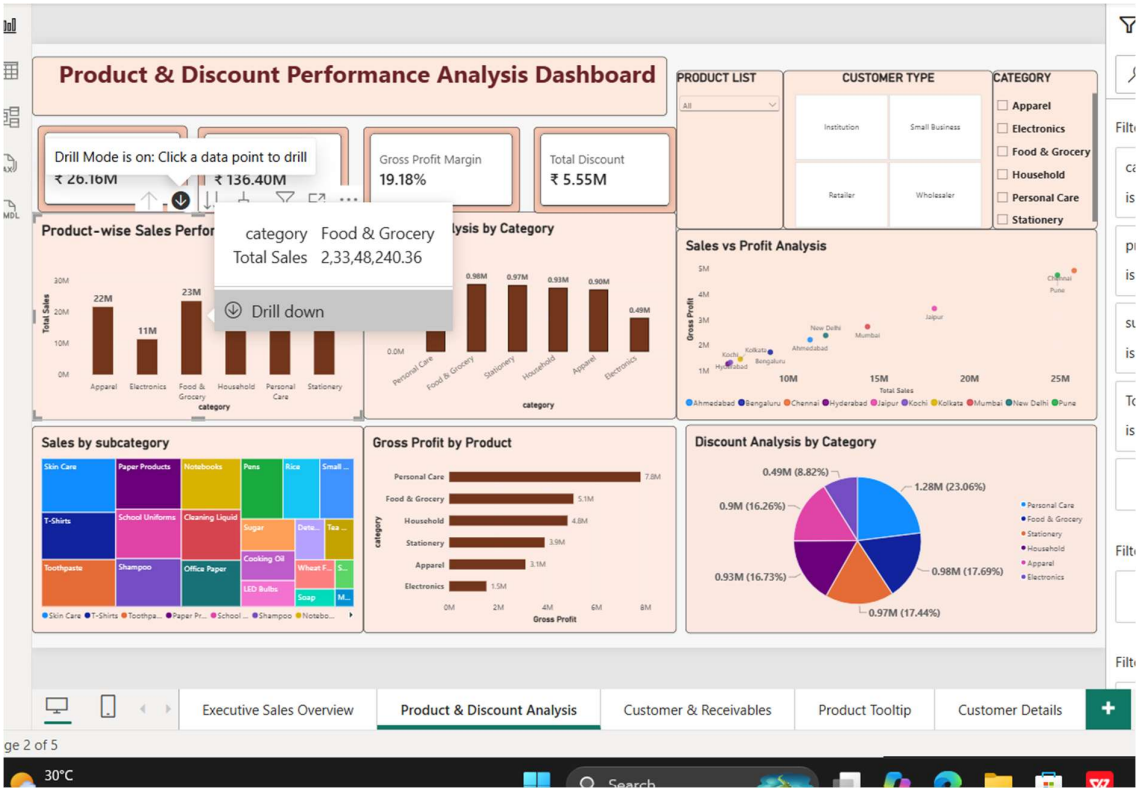
Category → Subcategory → Product

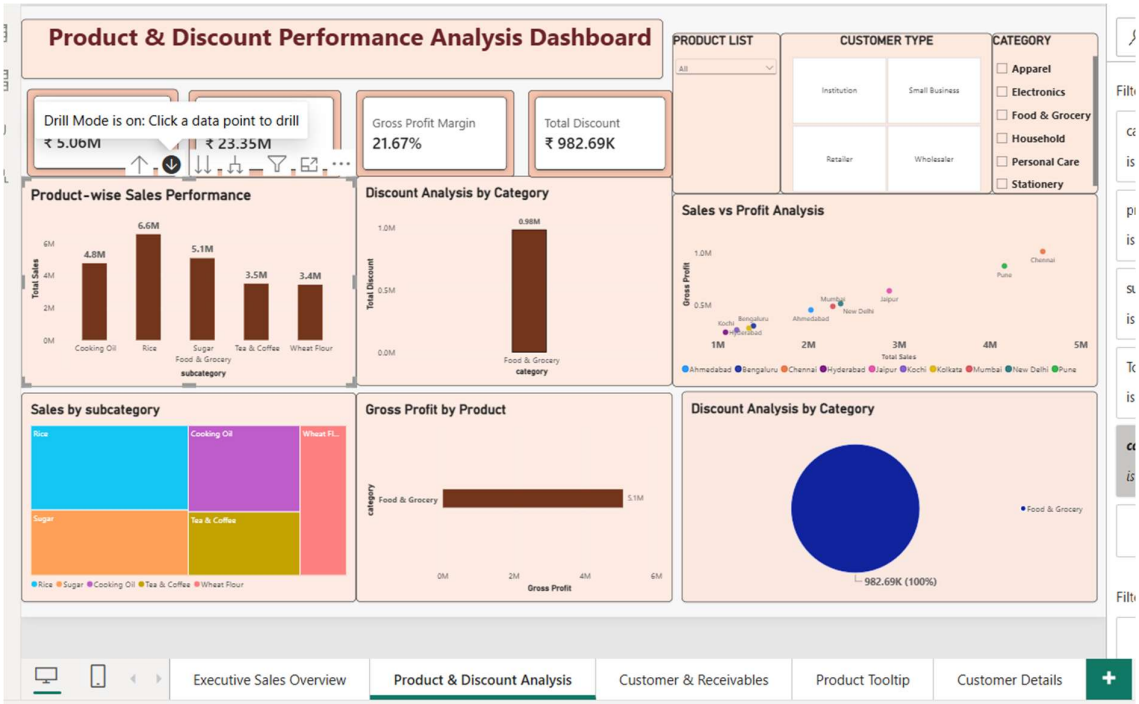
The product drill-down allows users to examine sales performance at different product levels.

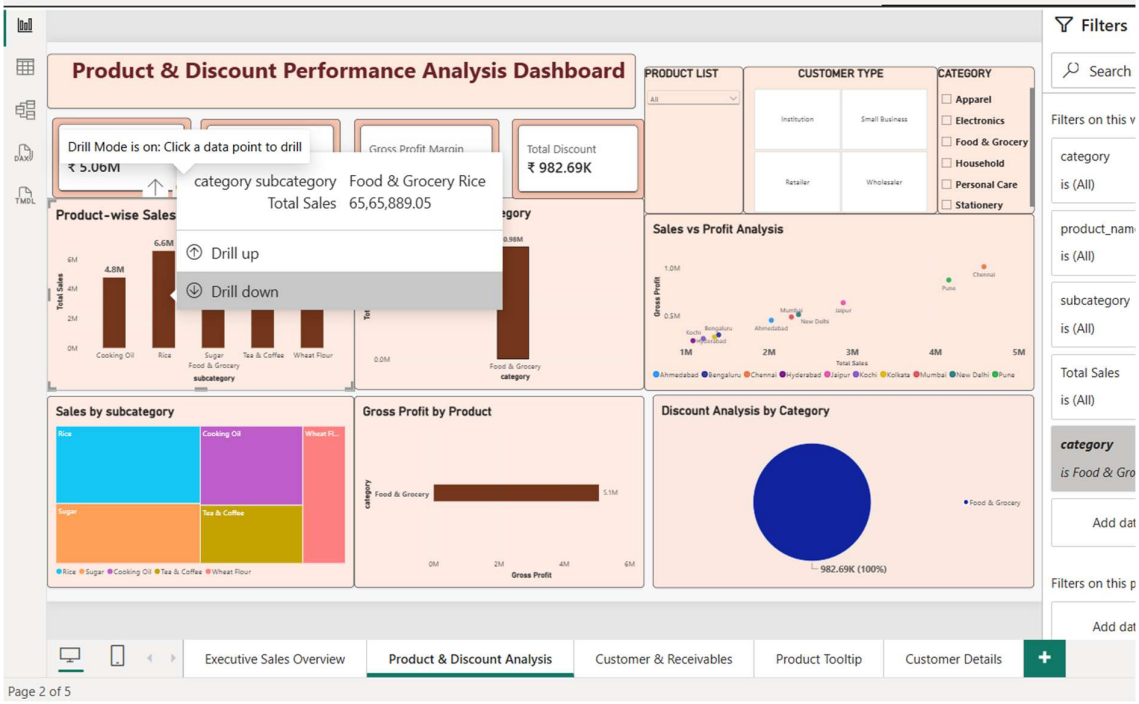
Users can begin with the overall category sales, move to subcategories and finally explore individual products.

For example, selecting a particular category allows users to examine its subcategories and the individual products contributing to its sales.

Drill-down helps users move from a summary-level analysis to detailed business information without creating separate charts for every level.




Product & Discount Performance Analysis Dashboard
Drill Mode is on: Click a data point to drill
₹ 5.06M
₹ 23.35M
Gross Profit Margin
21.67%
Total Discount
₹ 982.69K
PRODUCT LIST
CUSTOMER TYPE
Institution
Small Business
Retailer
Wholesaler
CATEGORY
Apparel
Electronics
Food & Grocery
Household
Personal Care
Stationery
Product-wise Sales Performance
Discount Analysis by Category
Sales vs Profit Analysis
Sales by subcategory
Gross Profit by Product
Discount Analysis by Category
982.69K (100%)
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details


Product & Discount Performance Analysis Dashboard
Drill Mode is on: Click a data point to drill
category subcategory Food & Grocery Rice
Total Sales 65,65,889.05
Drill up
Drill down
Total Discount
₹ 982.69K
Filters
Search
Filters on this v
category
is (All)
product_nam
is (All)
subcategory
is (All)
Total Sales
is (All)
category
is Food & Gro
Add dat
Filters on this p
Add dat
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details



Product & Discount Performance Analysis Dashboard
Gross Profit
₹ 1.31M
Drill up
Total Sales
₹ 6.57M
Gross Profit Margin
19.90%
Total Discount
₹ 280.17K
PRODUCT LIST
CUSTOMER TYPE
CATEGORY
Apparel
Electronics
Food & Grocery
Household
Personal Care
Stationery
Product-wise Sales Performance
Discount Analysis by Category
Sales vs Profit Analysis
Sales by subcategory
Gross Profit by Product
Discount Analysis by Category
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details

7.3 Drill-Through Functionality

Drill-through is used to navigate from one report page to another report page while carrying the selected data context.

In this project, a separate **Customer Details** drill-through page is created to examine customer-specific information.

When a user right-clicks a customer in a supporting customer visual and selects the Drill-through option, Power BI navigates to the Customer Details page with the selected customer's information.

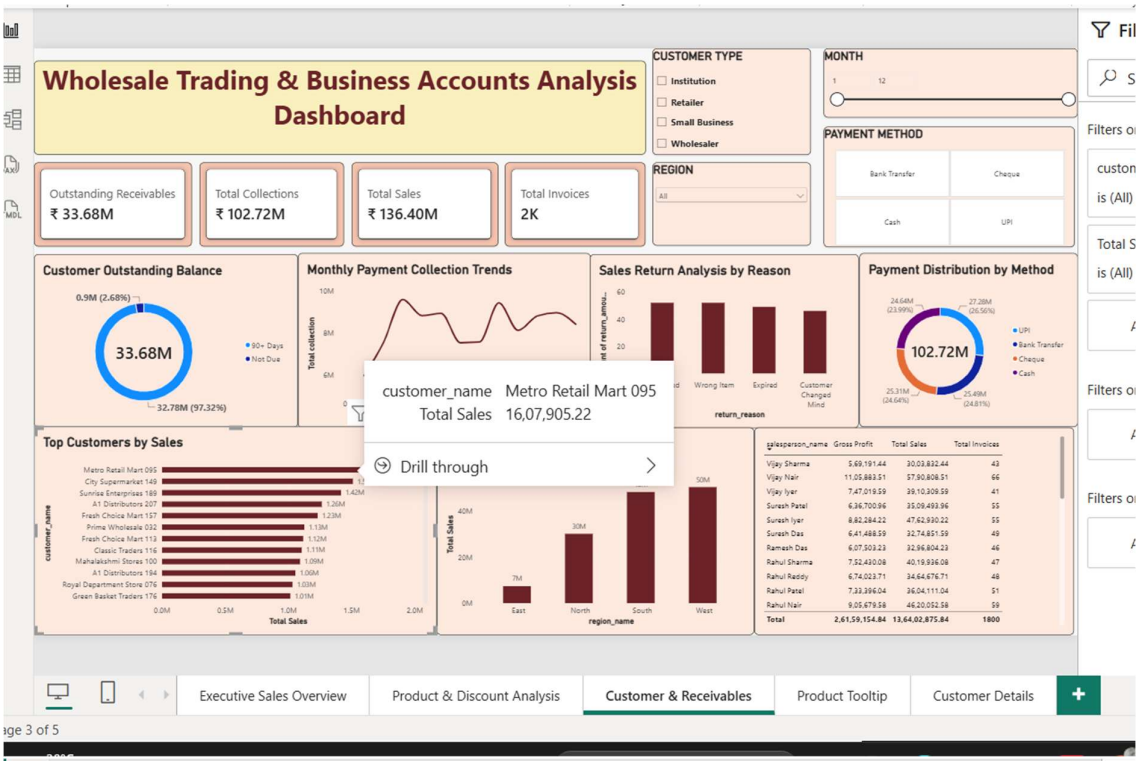
The Customer Details page includes:

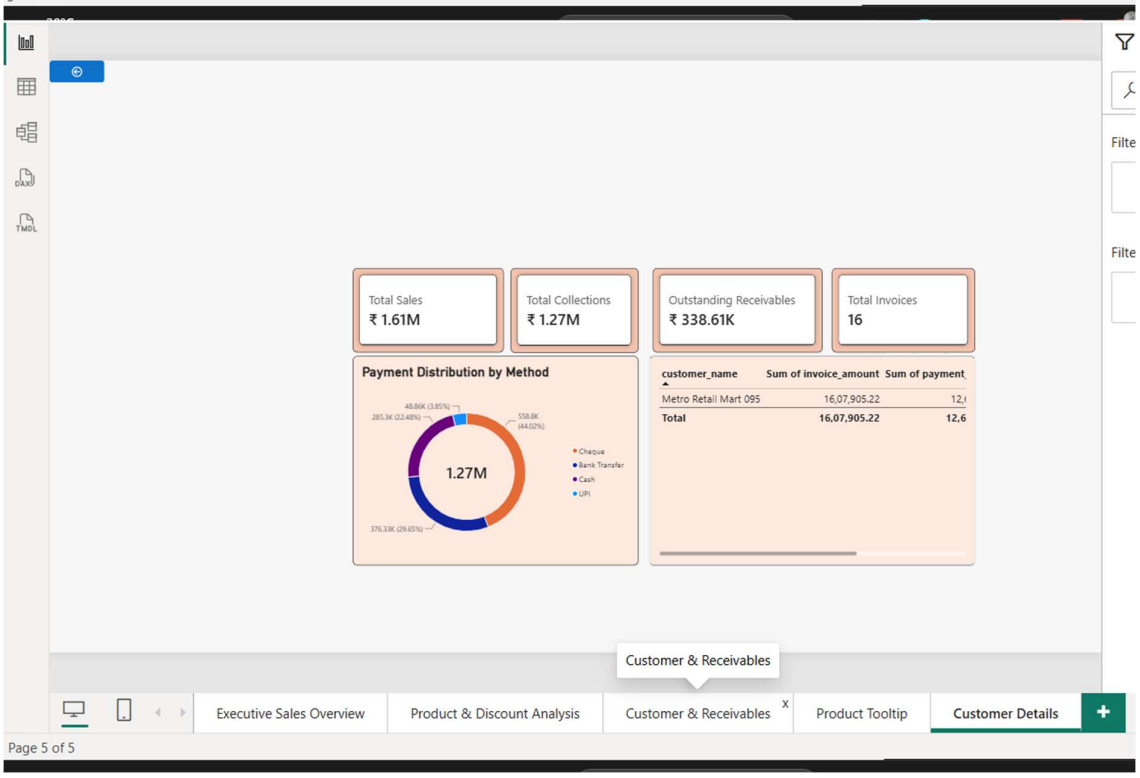
- Customer-wise total sales.
- Total collections received from the selected customer.
- Outstanding receivable amount.
- Invoice history showing customer transactions.
- Payment methods used by the selected customer.

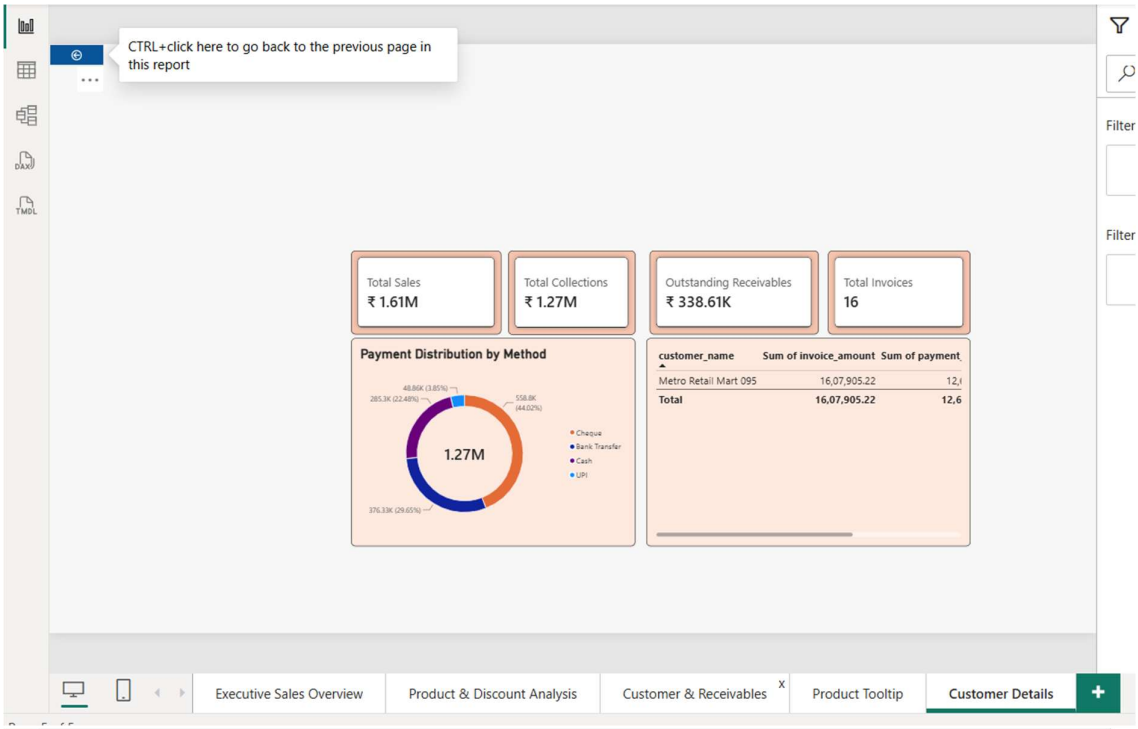
For example, selecting a customer from the customer sales visual allows management to examine that customer's sales contribution, payment history and pending balances in detail.

The Back button allows users to return to the previous report page.

This functionality helps management investigate individual customer accounts without manually searching through the complete transaction dataset.


Wholesale Trading & Business Accounts Analysis Dashboard
CUSTOMER TYPE
Institution
Retailer
Small Business
Wholesaler
MONTH
PAYMENT METHOD
Bank Transfer
Cheque
Cash
UPI
REGION
Outstanding Receivables
₹ 33.68M
Total Collections
₹ 102.72M
Total Sales
₹ 136.40M
Total Invoices
2K
Customer Outstanding Balance
33.68M
0.9M (2.68%)
32.78M (97.32%)
90+ Days
Not Due
Monthly Payment Collection Trends
Sales Return Analysis by Reason
Wrong Item
Expired
Customer Changed Mind
return_reason
Payment Distribution by Method
102.72M
UPI
Bank Transfer
Cheque
Cash
customer_name Metro Retail Mart 095
Total Sales 16,07,905.22
Drill through
Top Customers by Sales
Metro Retail Mart 095
City Supermarket 149
Sunrise Enterprises 189
A1 Distributors 207
Fresh Choice Mart 157
Prime Wholesale 032
Fresh Choice Mart 113
Classic Traders 116
Mahalakshmi Stores 100
A1 Distributors 194
Royal Department Store 076
Green Basket Traders 176
Total Sales
East
North
South
West
region_name
salesperson_name
Gross Profit
Total Sales
Total Invoices
Vijay Sharma 5,69,191.44 30,03,832.44 43
Vijay Nair 11,05,883.51 57,90,808.51 66
Vijay Iyer 7,47,019.59 39,10,309.59 41
Suresh Patel 6,36,700.96 35,09,493.96 55
Suresh Iyer 8,82,284.22 47,62,930.22 55
Suresh Das 6,41,488.59 32,74,851.59 49
Ramesh Das 6,07,503.23 32,96,804.23 46
Rahul Sharma 7,52,430.08 40,19,936.08 47
Rahul Reddy 6,74,023.71 34,64,676.71 48
Rahul Patel 7,33,396.04 36,04,111.04 51
Rahul Nair 9,05,679.58 46,20,052.58 59
Total 2,61,59,154.84 13,64,02,875.84 1800
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details
Page 3 of 5


Total Sales
₹ 1.61M
Total Collections
₹ 1.27M
Outstanding Receivables
₹ 338.61K
Total Invoices
16
Payment Distribution by Method
1.27M
Cheque
Bank Transfer
Cash
UPI
customer_name
Sum of invoice_amount
Sum of payment
Metro Retail Mart 095 16,07,905.22
Total 16,07,905.22
Customer & Receivables
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details
Page 5 of 5


CTRL+click here to go back to the previous page in this report
Total Sales
₹ 1.61M
Total Collections
₹ 1.27M
Outstanding Receivables
₹ 338.61K
Total Invoices
16
Payment Distribution by Method
1.27M
customer_name
Sum of invoice_amount
Sum of payment
Metro Retail Mart 095
16,07,905.22
Total
16,07,905.22
Filter
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details


Wholesale Trading & Business Accounts Analysis Dashboard
CUSTOMER TYPE
Institution
Retailer
Small Business
Wholesaler
MONTH
PAYMENT METHOD
Bank Transfer
Cheque
Cash
UPI
REGION
Outstanding Receivables
₹ 33.68M
Total Collections
₹ 102.72M
Total Sales
₹ 136.40M
Total Invoices
2K
Customer Outstanding Balance
33.68M
Monthly Payment Collection Trends
Sales Return Analysis by Reason
Damaged
Wrong Item
Expired
Customer Changed Mind
Payment Distribution by Method
102.72M
Top Customers by Sales
Geographical Sales Performance
East
North
South
West
salesperson_name
Gross Profit
Total Sales
Total Invoices
Vijay Sharma
Vijay Nair
Vijay Iyer
Suresh Patel
Suresh Iyer
Suresh Das
Ramesh Das
Rahul Sharma
Rahul Reddy
Rahul Patel
Rahul Nair
Total
Executive Sales Overview
Product & Discount Analysis
Customer & Receivables
Product Tooltip
Customer Details

7.4 Report-Page Tooltip

A report-page tooltip named **Product Tooltip** is created to display additional product-related information when users hover over the relevant product or category chart.

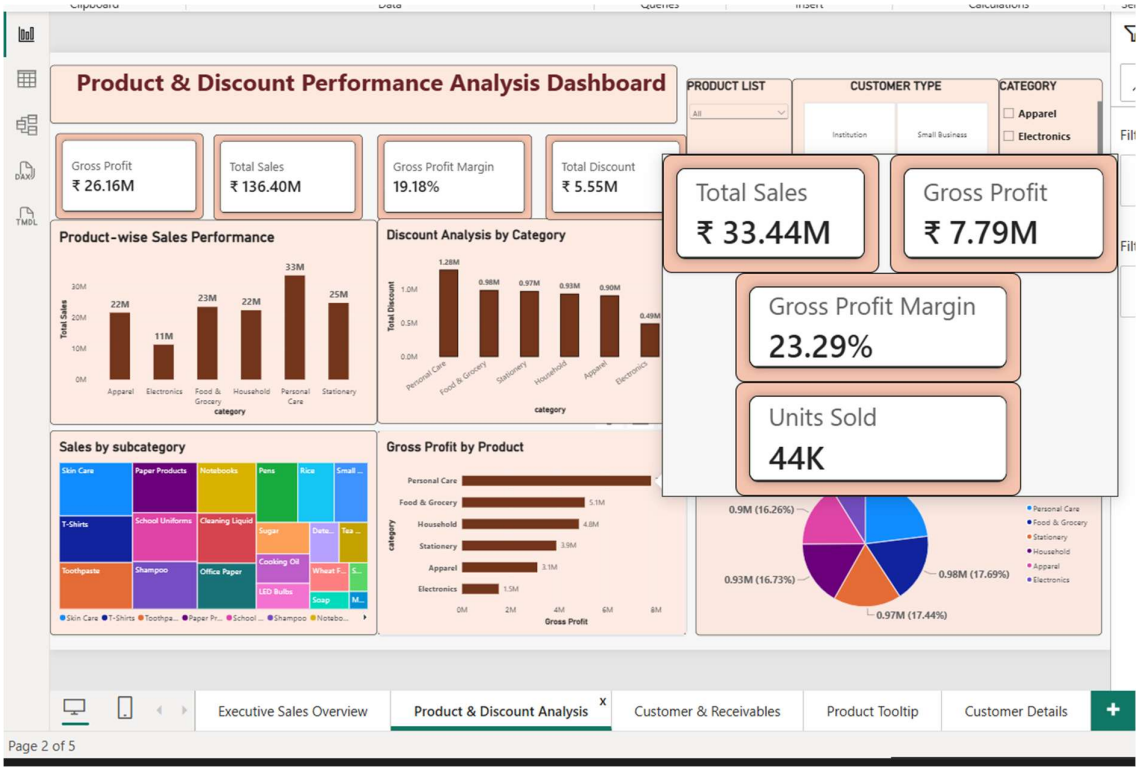
The tooltip contains four KPI indicators:

1. Total Sales
2. Gross Profit
3. Profit Margin
4. Units Sold

When users move the mouse pointer over a product or category in the assigned chart, the tooltip displays the corresponding financial information based on the hovered data point.

For example, hovering over a product allows users to view its sales contribution, gross profit, profit margin and units sold without navigating away from the existing report page.

This improves the user experience by providing additional information in a compact format and avoiding unnecessary visual clutter.



7.5 Visual-Level Filters

Visual-level filters are applied to individual visuals without affecting other visuals on the same page.

In this project, they can be used to display selected information within a particular chart or table.

For example:

- A Top N filter can display the top-performing customers based on sales.
- A product filter can restrict the gross profit chart to selected products.
- An ageing filter can display a particular overdue category.
- A return-reason filter can focus on specific return categories.

These filters help users examine specific information while keeping the remaining dashboard visuals unchanged.

7.6 Page-Level Filters

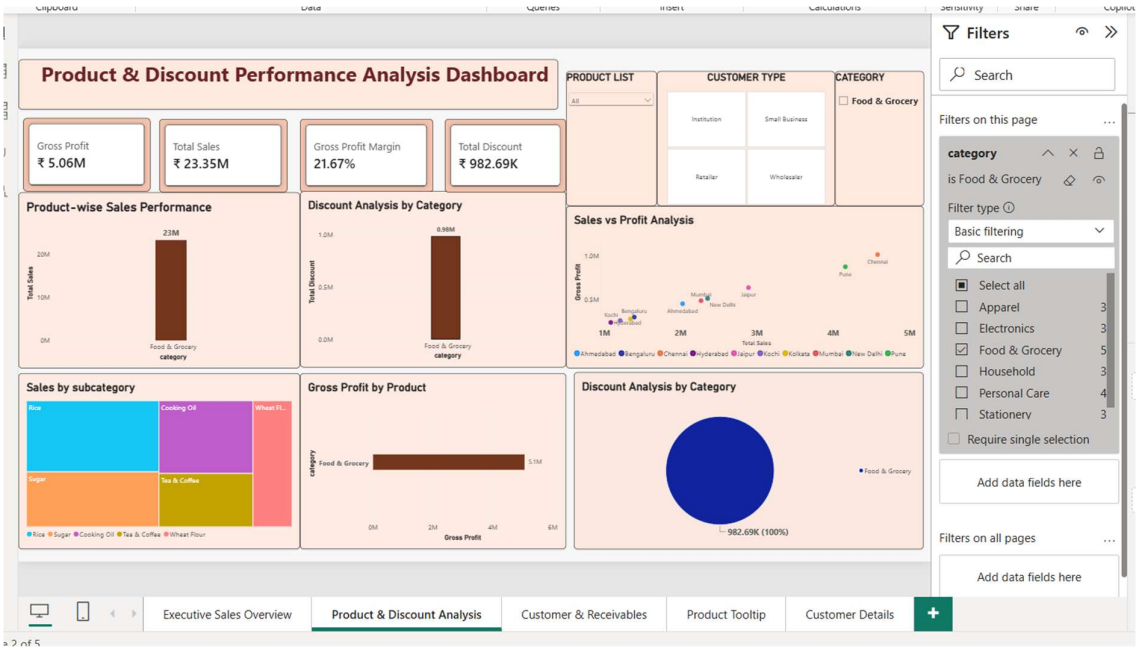
Page-level filters are applied to all relevant visuals within a particular report page.

In this project, page-level filters can be used to focus the analysis on a selected business area.

For example, applying a particular branch filter on the Branch & Salesperson Performance page allows the user to examine the sales and salesperson performance associated with that branch.

Similarly, applying a customer-related filter on the Customer & Receivables page helps focus the analysis on the selected customer group.

Page-level filters are useful when the purpose is to analyse a particular business aspect without affecting other report pages.



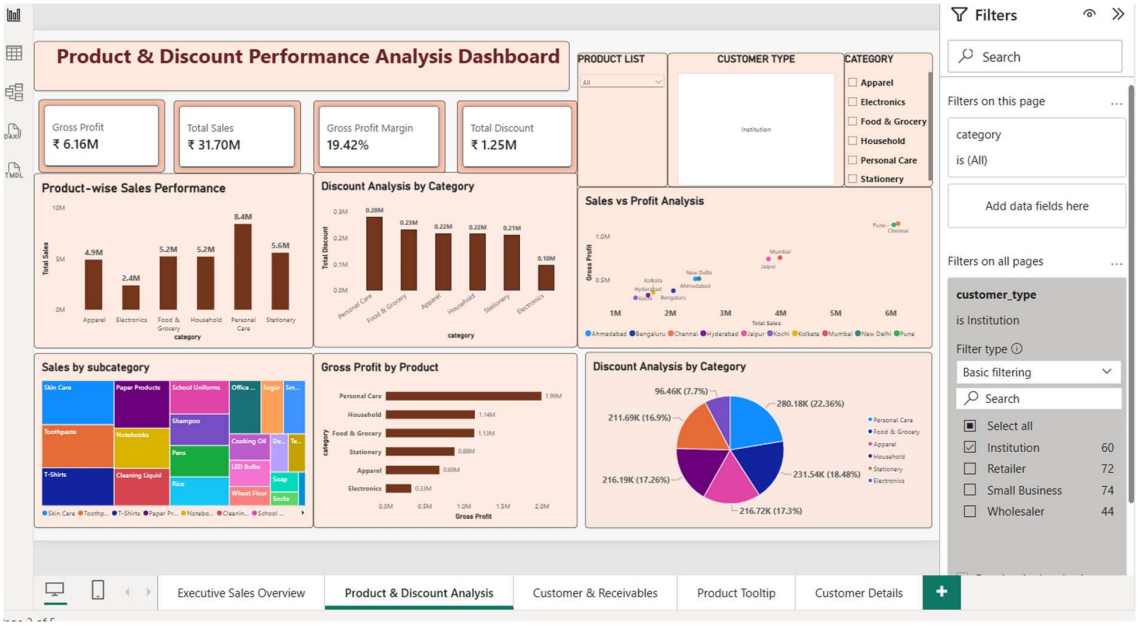
7.7 Report-Level Filters

Report-level filters are applied across all report pages and help maintain a common filtering condition throughout the dashboard.

In this project, a report-level Year filter can be used to restrict the entire report to the selected financial year.

For example, selecting FY 2024–2025 as the reporting period allows users to examine the relevant sales, collections, returns, customer and branch performance information across the report.

Report-level filters help maintain consistency in the analysis when users navigate between different report pages.





7.8 Edit Interactions

The Edit Interactions feature is used to control how one visual affects other visuals on the same report page.

For example, selecting a product category in the sales chart can filter or highlight the corresponding information in related visuals.

Similarly, selecting a particular customer in a customer sales visual can update other connected visuals to display the relevant customer information.

The interaction settings can be configured using Filter, Highlight or None, depending on the intended behaviour.

This functionality provides better control over the relationship between visuals and improves the overall dashboard experience.

7.9 Overall Dashboard Interactivity

The dashboard combines KPI cards, charts, tables, slicers, filters and navigation features to provide an interactive business analysis experience.

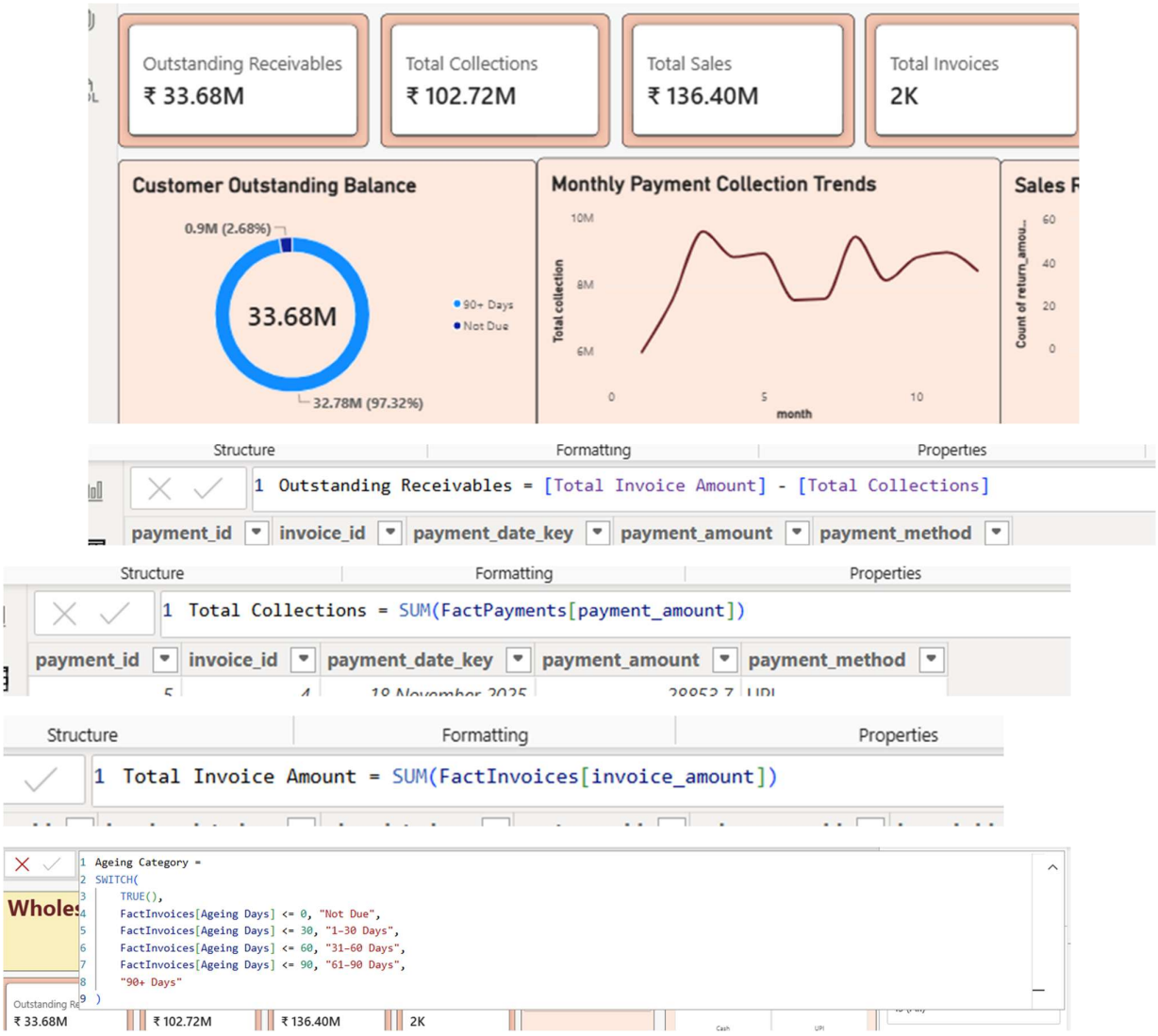
The different functionalities serve the following purposes:

- Slicers help filter the dashboard according to selected business categories.
- Drill-down helps explore information at different geographical and product levels.
- Drill-through provides detailed customer account information on a separate report page.
- Report-page tooltips display additional financial information when hovering over charts.
- Visual-level filters restrict information within individual visuals.
- Page-level filters apply conditions to all relevant visuals on a selected page.
- Report-level filters maintain common filtering conditions throughout the report.
- Edit Interactions controls how selections in one visual affect other visuals.

Together, these features allow users to move from an overall business summary to more detailed financial and operational analysis.

Receivables/Ageing Analysis Accuracy

• Calculated outstanding receivables by considering the invoice amount and payments received.
• Considered partial payments and identified the remaining unpaid balance.
• Classified invoices as overdue based on their due dates.
• Categorised overdue amounts into five ageing buckets: Not Due, 1–30 Days, 31–60 Days, 61–90 Days and 90+ Days.
• Used the customer-wise outstanding balance to identify customers with pending payments.
• Presented the overdue amounts through an ageing analysis chart to understand the duration of unpaid receivables.



The SWITCH function was used in the Ageing Category calculated column to classify outstanding invoices into different ageing buckets based on the number of overdue days.

It makes the DAX formula easier to read and manage by checking multiple conditions and assigning labels such as Not Due, 1–30, 31–60, 61–90 and 90+ Days.

8. Key Findings

1. Overall Sales and Collection Performance
The dashboard shows total sales of ₹136.40 million and total collections of ₹102.72 million. Outstanding receivables stand at ₹33.68 million, indicating that a portion of invoiced sales is yet to be collected.
2. High Concentration of Overdue Receivables
The Customer Outstanding Balance chart shows that ₹32.78 million (97.32%) of the displayed outstanding amount falls under the 90+ Days ageing category, while ₹0.90 million (2.68%) is shown as Not Due. This indicates that long-pending receivables form the major part of the displayed outstanding balance and require attention.
3. Product Sales and Profitability
In the displayed Product & Discount Analysis view, Rice-01 records approximately ₹6.57 million in sales and ₹1.31 million in gross profit, with a gross profit margin of 19.90%. The dashboard also shows total discounts of approximately ₹280.17 thousand in this view.
4. Regional Sales Performance
The geographical sales chart shows that the West region contributes approximately ₹50 million, followed by the South region at ₹48 million, the North region at ₹30 million and the East region at ₹7 million. The West and South regions are the leading sales contributors in the displayed view.
5. Customer Sales Contribution
The Top Customers by Sales chart highlights the contribution of individual customers. In the displayed view, Metro Retail Mart 095 leads with approximately ₹1.61 million in sales, followed by City Supermarket 149 at ₹1.51 million and Sunrise Enterprises 189 at ₹1.42 million. This helps identify key customers contributing to business revenue.
6. Collection Methods and Sales Returns
The payment distribution chart shows that UPI contributes approximately ₹27.28 million (26.56%) of the displayed collections, followed by Bank Transfer at ₹24.64 million (23.99%), Cheque at ₹25.31 million (24.64%) and Cash at ₹25.49 million (24.81%). The Sales Return Analysis by Reason chart also shows differences in return amounts across reasons, helping the business identify areas that may need attention.
The return-reason chart includes Damaged Packaging, Missing Item, Expired and Customer Changed Mind. The amounts vary across these reasons, indicating that return reasons should be monitored to identify recurring issues.

9. Business Recommendations

Based on the findings from the Power BI dashboard, the following five business recommendations are suggested:

1. Strengthen Receivables Collection
The business should prioritise the recovery of long-pending receivables, especially amounts falling under the 90+ days ageing category. Regular payment reminders and customer follow-ups can help reduce overdue balances.
2. Focus on Profitable Product Sales
The business should monitor product-wise gross profit and profit margins rather than focusing only on sales volume. Products generating lower margins should be reviewed to control costs and improve profitability.
3. Improve Regional Sales Performance
The business should study the sales strategies followed in the West and South regions, which contribute higher sales in the displayed dashboard. Similar approaches can be considered to improve performance in lower-contributing regions, such as the East.
4. Strengthen Relationships with Key Customers
The business should maintain strong relationships with high-contributing customers through regular communication, timely order fulfilment and better customer service. It should also monitor their payment behaviour to manage credit risk.
5. Monitor Payment Methods and Product Returns
The business should continue supporting commonly used payment methods such as UPI, bank transfer, cheque and cash. It should also regularly analyse product return reasons and take corrective action to reduce recurring issues such as damaged packaging, missing items and expired products.

10. Conclusion

Through this project, I learned how Microsoft Power BI can be used to convert wholesale business transaction data into an interactive dashboard for financial and operational analysis.

I explored how KPI cards, column charts, bar charts, treemaps, doughnut charts, line charts and tables can be used to present different aspects of business performance.

The dashboard provides a framework for analysing sales revenue, gross profit, discounts, customer collections, outstanding receivables, overdue invoices, sales returns and regional performance.

This project also helped me understand the practical application of Power BI functionalities such as slicers, filters, drill-down, drill-through, report-page tooltips and Edit Interactions. These features allow users to explore information at different levels and examine individual business transactions more effectively.

From a commerce and FinTech perspective, this project helped me understand how business intelligence tools can support accounting analysis, receivables monitoring, profitability evaluation and management decision-making.

Overall, this project provided practical exposure to financial data analysis and data visualisation using Microsoft Power BI. It strengthened my understanding of how accounting information can be transformed into meaningful business insights.

11. Assumptions and Methodology

The following assumptions are considered while interpreting the dashboard:

1. The analysis is based on the synthetic wholesale trading dataset provided for the academic project.
2. All financial amounts are expressed in Indian Rupees (₹).
3. Net sales are calculated after considering applicable discounts.
4. Gross profit is calculated as net sales less the cost of goods sold.
5. Total collections are calculated using recorded customer payment transactions.
6. Outstanding receivables are calculated at invoice level after considering payments received and accepted return or credit adjustments, wherever applicable.
7. An invoice is considered overdue when its outstanding balance is greater than zero and its due date is earlier than the selected as-of date.
8. Overdue ageing is classified into Not Due, 1–30, 31–60, 61–90 and 90+ days.
9. The selected as-of date for receivables and overdue ageing is [Insert Date].
10. Payment and return analysis is based on their respective transaction dates and applicable date relationships.
11. The report covers FY 2024–2025 and the dataset is intended for educational analysis only.